MIL-D-37005A
29 September 1976
SUPERSEDING
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5 September 1973

MILITARY SPECIFICATION

DISK, FILTERING, MICROPOROUS, Hydrosol Type, 47 mm. Diameter, 100s

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a hydrosol type, white microporous membrane filter disk suitable for quantitative surface retention of bacteria suspended in liquids.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Boxes, Shipping, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Hq, Defense Personnel Support Center, ATTN: Directorate of Medical Materiel, DPSC-ATT, 2800 S. 20th St., Phila. PA 19101, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	Marking for Shipment and Storage.

(Copies of specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposals shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM F316	Pore Size Characteristics of Membrane Filters for Use with Aerospace Fluids.

(Application for copies should be addressed to American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HANDBOOK OF CHEMICAL MICROSCOPY

Volume I, Third Edition

(Application for copies should be addressed to John Wiley and Sons, 605 Third Avenue, New York, NY 10016.)

3. REQUIREMENTS

3.1 Material. Material shall be suitable for the purpose intended and as hereinafter specified.

3.1.1 Filter disks. Filter disks shall be fabricated of white cellulose ester polymer sheet which has a nominal pore size of .45 microns.

3.1.2 Pads. Pads shall be fabricated of white cellulose.

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3.2 Construction.

3.2.1 Filter disks.

3.2.1.1 Grid marking. A grid of squares shall be imprinted on the surface of the filter disks. Each side of each square shall be 3.1 ± 0.1 millimeters (mm.) long. The length of a side shall be the distance between two intersecting sides plus the width of one intersecting line. Grids shall extend throughout the surface, and grid marking shall be legible and permanent when tested as specified in 4.4.8.

The area of each square is 1/100 of the effective filtering area of 9.6 square centimeters (sq. cm.) when disk is used with microporous filtering disk holder (see 6.2).

3.2.1.2 Thickness. Thickness shall be 150 ± 20 microns.

3.2.1.3 Porosity. At least 77% of the filter disk shall be pores when tested as specified in 4.4.1.

3.2.1.4 Refractive index. Refractive index of filter disk shall be 1.51 ± 0.01 when tested as specified in 4.4.2.

3.2.2 Filter pads.

3.2.2.1 Dimensions. Size of filter pads shall be such that they absorb the amount of nutrient specified in 3.2.2.2 and so that they are suitable for use with filter disk. Diameter shall be 47 mm. (nominal) and thickness shall be 1/32 inch (in.) (0.8mm) nominal.

3.2.2.2 Absorptivity. Filter pads shall absorb 1.5 ± .2 milliliters (ml.) when tested as specified in 4.4.3.

3.3 Performance.

3.3.1 Bubble point. Bubble point pressure shall be 32 pounds per square inch (p.s.i.) ± 2 p.s.i. when tested as specified in 4.4.4.

3.3.2 Diffusibility. Water shall diffuse uniformly through the filter disk in 15 seconds, and there shall be no dry spots when tested as specified in 4.4.5.

3.3.3 Flow rate. Flow rate shall be at least 55 ml./minute/sq. cm. when tested as specified in 4.4.6.

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3.3.4 Organism retention. Serratia marcescens shall be retained by the filter disk when tested as specified in 4.4.7.

3.3.5 Toxicity. Filter disk shall not inhibit bacterial growth when tested as specified in 4.4.8.

3.4 Immediate package. Filter disks and pads shall be wrapped (10 filter disks and 10 filter pads) to form an immediate package. Ten such immediate packages shall constitute one unit of issue. Immediate package shall have an indicator which shows when sterilization is complete. Each unit of issue and immediate package shall have instruction for interpreting the indicator.

3.4.1 Autoclavability. Immediate package,disks, pads and wrapping paper shall not deteriorate, buckle or deform and indicator shall perform as shown on instructions when tested as specified in 4.4.9. Diameter of disk shall be 46.6 ± 0.6mm after autoclaving (4.4.9). Disk shall be measured between 10 and 15 minutes after removal from the autoclave. Temperature and humidity for waiting period and measurement shall be 25 degrees Celsius (°C) and 40% relative humidity.

3.5 Workmanship. Workmanship shall be first class throughout. Immediate package, filter disks and pads shall be free of defects listed in Table III and other defects which detract from their appearance or may impair their serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Records. Records of examinations and tests performed by or for the contractor shall be maintained by the contractor and made available to the Government, upon the Government's request, at any time, or from time to time, during the performance of the contract and for a period of 3 years after delivery of the supplies to which records relate.

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4.1.2 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties) of the item.

4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105 and Table I.

Table I. For examination

	Inspection level	AQL * (percent defective)
For visual examination		
Major A		
Immediate package	S-1	2.5
Major B		
Disk	S-1	2.5
Major C		
Filter pad	S-1	2.5
For dimensional examination	S-1	2.5

* If acceptance number for any category is exceeded, the lot may be re-examined for approval in that category at the same AQL, using inspection level II.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with MIL-STD-105 and Table II. Acceptance number shall be zero.

Table II For tests

Component (unit of product)	Characteristic	Requirement	Test procedure	Insp. level
Disk	Porosity	3.2.1.3	4.4.1	S-1
Disk	Refractive index	3.2.1.4	4.4.2	S-1
Filter pad	Absorptivity	3.2.2.2	4.4.3	S-1
Disk	Bubble point	3.3.1	4.4.4	S-1
Disk	Diffusibility	3.3.2	4.4.5	S-1
Disk	Flow rate	3.3.3	4.4.6	S-1
Disk	Organism retention	3.3.4	4.4.7	S-1
Disk	Toxicity	3.3.5	4.4.8	see 4.4.8
Disk	Permanency of marking	3.2.1.1	4.4.8	see 4.4.8
Immediate package	Autoclavability	3.4.1	4.4.9	S-1

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4.3 Examination. The immediate package and components shall be examined to determine compliance with all requirements contained in this specification. Examination shall be in accordance with, but shall not be limited to, the defects listed in Table III.

Table III. Classification of defects

Categories	Defects
Major A	
101	Component missing.
102	Indicator missing.
103	Instructions for indicator not complete, correct or legible.
104	Package not free of tears.
Major B	
111	Disk not free of tears.
112	Disk not free of wrinkles.
113	Grid markings not complete, correct or legible.
Major C	
121	Pad not free of tears.
122	Pad not free of wrinkles.

4.3.1 Dimensional examination. The filter disk shall be examined for defects in dimensions. Any dimension not within the tolerance specified herein shall be classified as a defect.

4.4 Tests. Tests shall be conducted to determine compliance with specification requirements.

4.4.1 Porosity. Porosity shall be determined by one of the following methods:

a. Calculate the theoretical weight of a solid filter by multiplying the volume by the density of the cellulose ester polymer and determine the actual weight of the filter disk.

$$\frac{\text{Theoretical weight of a solid filter disk} - \text{Actual weight of filter}}{\text{Theoretical weight of a solid filter disk}} \times 100 = \text{percent porosity}$$

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b. Calculate the volume of the sample filter disk. Introduce mercury into the pores of the filter disk.

$$\frac{\text{Volume of mercury introduced}}{\text{Theoretical volume of filter disk}} \times 100 = \text{percent porosity}$$

4.4.2 Refractive index. The following test shall be performed:

Cut a 1 mm. by 5 mm. strip from a filter disk. Dimensions shall be approximate, and cutting shall be done with a sharp instrument so that smooth edges are obtained. Place strip on a glass slide. Add a drop of oil to the strip and cover strip with a cover glass. Place the prepared slide on the stage of a microscope (magnification 100X) and position so that the microscope is focused on an edge of the strip (interface of the oil and strip). If the refractive indices of the oil and strip are different, a line called a Becke line will appear at the interface. Move the objective lens up or down so that the Becke line is moving from the strip to the oil. Upward movement of the lens indicates that the refractive index of the oil is greater than that of the strip. Downward movement of the lens indicates that the refractive index of the strip is greater than that of the oil. If the strip and Becke line cannot be brought into focus, the refractive index of the strip and the oil are the same. All oils used shall be within the tolerances of 3.2.1.4, and each disk shall be tested with at least two oils, one of greater and one of lesser refractive index that the disk. (Theory of the refractive index test is given in the HANDBOOK OF CHEMICAL MICROSCOPY.)

4.4.3 Absorptivity. The following test shall be performed:

Place pad in a container which has a diameter slightly larger than that of the pad. Slowly add 1.7 ml. of endo broth to the center of the pad and allow to stand for 10 minutes (see 6.4). If pad does not absorb all of the endo broth, repeat the above procedure with another pad using 1.3 ml. of endo broth. If pad does absorb the entire 1.7 ml. of endo broth, repeat the above procedure with another pad using 1.8 ml. of endo broth (see 6.3).

4.4.4 Bubble point. Bubble point test shall be run in accordance with ASTM Designation F-316, using water as the test liquid.

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4.4.5 Diffusibility. Filter disk shall be dropped flat on water. Disk shall be observed for dry spots and diffusion through the disk shall be timed with a stopwatch. (Wet areas become darker while dry areas do not change in color intensity).

4.4.6 Water flow rates. Water flow shall be determined at 25°C. at a pressure differential of 70 cm. of mercury.

4.4.7 Organism retention. Test dilution shall be a phosphate-buffered or a saline suspension of 1.0×10^3 cells of a 24-hour culture of Serratia marcescens per ml. All equipment shall be sterile. Using forceps, a sterile filter disk that has been autoclaved twice in accordance with 4.4.9 shall be placed on the filter support base and the funnel unit shall be placed on the disk and secured by means of the appropriate attachment device. Test dilution shall be drawn through the test disk by applying a vacuum to the sterile flask. The disk shall then be rinsed three times with 20 ml. of sterile dilution water. A sterile filter holder shall then be attached to another filtration flask and, using an aseptic technique, the filtrate from the test disk shall be passed through a reference disk. This reference disk shall be the subject disk from a lot which has been furnished to the Government and which meets the requirements of this specification; or reference disk shall be fabricated from the same material but shall have a smaller pore size than the subject item. After filtration, reference disk shall be placed on Wilson's Peptone Agar at 32°C. for 24 hours (see 6.5). A zero count on the incubated referenced disk indicates sterility of the filtrate and retention by the test filter disk.

4.4.8 Toxicity. Test dilution shall be a phosphate-buffered suspension of 30 to 80 viable cells of a 24-hour culture of E. coli per 0.1 ml.

Filter disk shall be placed on the filter base, and funnel unit shall be placed on the disk and made secure by an appropriate attachment device. Twenty ± 2 ml. of sterile dilution water shall be added to funnel. 0.1 ml. of E. coli dilution shall be pipetted into the funnel. Dilution shall be drawn through the disk by means of a vacuum, and disks shall be placed on membrane (m) Plate Count Broth. With the same pipet, 0.1 ml. of E. coli dilution shall be delivered

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to the surface of 100 mm. plate of Plate Count Agar. E coli dilution shall be spread over the entire surface of the plate with a sterile glass streaking rod so that the dilution is evenly distributed. Incubate the inverted disk and agar plates at 35° C. ± 5° for 24 hours and count the resultant colonies.

Five disks and five agar plates shall be run. The disks shall be considered non-toxic if the arithmetic mean of the count on the five disks is at least 90% of the arithmetic mean of the count on the five agar plates. The colonies on the disks shall measure at least 0.5 mm. in diameter. If the disks do not meet the above criteria of 90% growth, toxicity testing procedures may be repeated with five more disks. If test is repeated, arithmetic mean shall be calculated by comparison with 10 (5 + 5) disks, with 10 (5 + 5) agar plates. If disks fail to meet the criteria of 90% growth on 10 disks, lot shall be rejected (see 6.6).

Grid lines shall be observed to determine conformance with 3.2.1.1.

4.4.9 Autoclaving. Immediate package shall be autoclaved for 10 minutes at 121°C. and 15 pounds per square inch pressure.

* 4.5 Examination for preparation for delivery. An examination shall be made to determine whether the packaging, packing, and marking comply with the requirements of Section 5. Defects shall be scored as specified in Table IV. Sampling shall be in accordance with MIL-STD-105. The sample unit shall be one container fully prepared for delivery. The lot shall be the number of containers offered for inspection at one time. The inspection level shall be S-2 and the acceptable quality level shall be 4.0 defects per hundred units.

* Table IV. Examination for preparation for delivery

Examination	Defect
Containers	Not as specified.
Quantity	Quantity in interior packages and shipping containers not as specified.
Packaging and packing materials	Any component missing or damaged.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, tape improperly applied, or containers distorted.
Markings	Omitted, incorrect, illegible, improper size, location, sequence, or method of application.

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* 5. PREPARATION FOR DELIVERY

5.1 Packaging. Packaging shall be Level A or C as specified (see 6.1).

5.1.1 Unit of issue. One package containing one hundred disks, as specified, constitutes one unit of issue.

5.1.2 Level A.

5.1.2.1 Unit package. Ten immediate containers, containing filters and pads as specified in 3.4, shall be packaged in a snug-fitting box constructed in accordance with PPP-B-566 or PPP-B-676. Closure shall be adequate to prevent accidental opening. As an alternate, the unit package may be a polystyrene box with a friction closing lid.

5.1.2.2 Intermediate package. Twelve unit packages shall be packaged in a snug-fitting box constructed in accordance with PPP-B-566, PPP-B-636, type CF, class domestic or PPP-B-676. Closure shall be adequate to prevent accidental opening.

5.1.3 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing. Packing shall be Level A, B or C as specified (see 6.1).

5.2.1 Level A. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 3, style 3; PPP-B-601, overseas type; PPP-B-621, class 2; or PPP-B-636, class weather-resistant. Closure and strapping shall be as specified in the appendix of the applicable box specification.

5.2.1.1 Case liner. Each Level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes. Fiberboard box shall be waterproof in accordance with paragraph 30.4 of PPP-B-636.

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5.2.2 Level B. Twelve intermediate packages (144 units) shall be packed in an exterior container designed for a type 2 load and constructed in accordance with PPP-B-585, class 1, style 3; PPP-B-601, domestic type; PPP-B-621, class 1; or PPP-B-636, type CF, class domestic. Closure shall be as specified in the appendix of the applicable box specification.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. Contract or purchase order number and name of contractor shall be shown.

5.3.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number and date of this specification.
b. National Stock Number (NSN).
c. Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

6.2 Holder referred to in 3.2.1.1 is NSN 6640-00-299-8691, HOLDER, MICROPOROUS FILTERING DISK, in accordance with type I of MIL-H-36376 (DSA-DM).

6.3 This specification covers the following item listed in the Federal Supply Catalog:

National Stock No.	Item Identification
6640-00-299-8692	DISK, FILTERING, MICROPOROUS, Hydrosol Type, 47 mm. Diameter, 100s

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6.4 Endo broth referred to in 4.4.3 is NSN 6505-00-687-8436, ENDO BROTH, Membrane Filter, 2 cc, 24s.

* 6.5 Wilson's Peptone Agar may be obtained from Baltimore Biological Laboratory, Inc., Baltimore, MD (see 4.4.7).

* 6.6 The m Plate Count Broth is available from the Difco Co., Detroit, MI (see 4.4.8).

6.7 This specification does not cover all types, classes, grades or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the type normally procured to meet military requirements.

6.8 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Custodians:
Army - MD
Navy - MS
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Preparing activity:
Defense Supply Agency - DM

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DOCUMENT IDENTIFIER AND TITLE MIL-D-37005A - DISK, FILTERING, MICROPOROUS, Hydrosol Type, 47 mm. Diameter, 100s

NAME OF ORGANIZATION AND ADDRESS	CONTRACT NUMBER
	MATERIAL PROCURED UNDER A ☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?
   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*	TELEPHONE NO.
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